Quiz — 1.2.1 Operating Systems — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Section A (8 marks — 1 each) 1. **B** — virtual memory uses secondary storage (swap/page file) as if it were RAM; it does not add physical RAM. 2. **B** — pages and frames are fixed-size; variable blocks describe segmentation. 3. **B** — variable-size segments leave gaps → external fragmentation (paging → internal). 4. **B** — registers (PC, ACC) are pushed onto a stack so the interrupted task can resume. 5. **B** — an RTOS guarantees a response within a deadline (e.g. ABS, pacemaker). 6. **C** — round robin uses a fixed quantum per process in turn. 7. **B** — the BIOS (firmware on ROM) runs first, performs POST, then loads the OS. 8. **B** — source compiled to platform-independent intermediate code (bytecode) run by a process VM (JVM).

Section B (12 marks)

9. **[2]** Any two (1 each): paging uses **fixed-size** blocks, segmentation **variable-size**; paging divided by OS/physical memory, segmentation by the program's **logical structure**; paging suffers **internal** fragmentation, segmentation **external**.

10. **[2]** Disk thrashing = excessive paging/swapping of pages between RAM and disk (1); so much time is spent moving pages that little useful processing happens and the machine slows to a crawl (1).

11. **[3]** Any three: CPU checks the interrupt register at the **end of each FDE cycle** (1); finishes the current instruction / compares priorities (1); **pushes the registers (PC, ACC) onto the stack** (1); loads the address of and runs the correct **ISR** (1); **pops registers off the stack** to resume the task (1) — stack is **LIFO** so nested interrupts resume in the correct order (1). Max 3.

12. **[2]** Washing machine → **embedded OS** (one dedicated task, limited resources, stored in ROM, reliable/efficient) (1). Mainframe → **multi-user OS** (many users share resources via scheduler + permissions) (1).

13. **[3]** A device driver is software that lets the OS control/communicate with a specific hardware device (1), translating OS commands into instructions the device understands (1); it is hardware-specific because each device has its own commands and OS-specific because it must use that OS's interfaces/conventions (1).

Section C (5 marks) 14. **[5]** Up to 5 from: a **virtual machine** is software that emulates a complete computer (1); a **system VM** can run a whole guest OS inside the host (1), so the company can install several guest OSes on one physical machine and test the software on each (1). Advantage: saves cost / no extra hardware needed, and provides a safe **sandbox** so untrusted software cannot harm the host (1). Drawback: the VM runs **slower than native** hardware / the VM consumes host resources (RAM, CPU) (1).

Total: 8 + 12 + 5 = 25